A

All important questions

Bt102.....microbiology

Final term spring 2020

By zareen warraich.....future zoologist group.

1. Define and type of algae. 5 marks

Define :

The definition of algae is a single or multi-cellular organism that has no roots, stems or leaves and is often found in water

These are simple eukaryotic cells. Some are unicellular, others are multicellular (thallus); however, they lack tissues such as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are mostly photoautotrophs; however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the face of the earth.

Types of algae:

1. **Phaeophyta (Kelp):** Commonly known as brown algae, brownish in color, contains cellulose and alginic acid cell walls, are multicellular and contain chlorophyll *a* and *c*, xanthophylls. These store carbohydrates and are used for algin which is a thickener in ice creams. Can grow upto 20 cm in one day.

2. **Rhodophyta:** Red Alage, these are reddish in color and contain cellulose in their cell walls. Most are multicellular and contain chlorophyll a and d, and phycobiliproteins. They store glucose polymer and used for agar and carrageenan (both are used as thickeners of foods).

3. **Chlorophyta:** Green Algae: have all the features of red algae except that chlorophyll is of a and b types only. These algae are believed to have given rise to terrestrial plants.

4. **Bacillariophyta:** Cell wall is composed of pectin and silica. They are unicellular and possess chlorophyll a and c, and carotene xanthophylls. These algae store oil and produce domoic acid with which human can be intoxicated by eating contaminated mussels. **Diatoms**: Come in beautiful colors and shapes.

5. **Dinoflagellata:** These are unicellular algae collectively called as plankton (free floating organisms). These cells have cellulose in their cytoplasmic membrane which gives the membrane rigidity. They have chlorophyll *a* and *c*, and carotene. These algae store starch in them. Dinoflagellates in the genus Alexandrium produce neurotoxins (called saxitoxins) which can cause paralytic shellfish poisoning in humans when they consume contaminated mussels and clams. Blooms of Alexndrium gives ocean a deep red color commonly known as red tide.

6. **Oomycota (Water molds):** They are basically decomposers of dead matter. Superficially, these resemble with fungi; however, these are motile with flagella and their cell wall is composed of cellulose not chitin. Their spores are called zoospores. Terrestrial members of water molds are also plant parasite. *Phytophthora infestans* was responsible for Irish potato blight during mid-1800s. *P. ramorum* causes sudden oak death

2. HYPERSENSITIVITY AND ITS TYPE 5 marks

What is hypersensitivity?

It is a normal Immune response but in a damaging way. In other words, immune system is acting in a way for which it has made to. So when an antigen gets into the body, immune system recognizes it as foreign and start making antibodies against the antigen. However, the consequences are bad and damaging to the body. It may be called an abnormal antigenic response. Allergy is another name for hypersensitivity. Those antigens that act in a bad way are called allergens **Types of hypersensitivity:**

Hypersensitivities can be grouped into 4 types based on the types of antibodies or cells involved.

1. **Type I (Anaphylactic):** Also called anaphylaxis or immediate hypersensitivity. This is IgE mediated.
2. **Type II (Cytotoxic):** IgG and IgM mediated as these antibodies attach to cells and destroy them by activating complement.
3. **Type III (Immune complex mediated):** Immune complexes are formed by IgM or IgG antibodies and these complexes lodge in the capillary beds and cause damage by complement activation.
4. **Type IV (delayed type cell-mediated):** CD4+ helper cells along with macrophages are involved.

3. Define antigen and epitope

1. **Antigen**: A foreign substance that provokes immunity is called an antigen.
   The definition of antigen is a harmful substance which enters the body which causes the body to make antibodies as a response to fight off disease.
   More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably. o Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens. o Antigens are foreign to the body. o **Degradability**: Inert molecules are not antigenic.
2. **Epitopes**: Specific regions on an antigen. In other words, any antigen has multiple regions each one of those act as a small antigen. These are called epitopes.These epitopes are also called as antigenic determinants. o Most antigens have MW >10,000 D. Molecules that are less than 10,000 D are generally poor antigen.

   A small molecular region of an antigen that binds to a particular antibody or antigen receptor on a T cell; an antigenic determinant. A single antigen can have multiple epitopes

4. Describe two transported system in the blood and why it is necessary 10 marks

Cardiovascular and **lymphatics** are the two transport systems of the body by which cells of the immune system patrol the whole body.

Cardiovascular system consists of the heart and associated blood vessels (arteries and veins) in which blood circulates lymphatics are channels that start from tissue spaces and collect tissue fluid (called lymph) from tissue spaces and bring them back to the heart. Along these lymphatic channels, there are strategically placed lymph nodes which filter lymph for microbes much like blood is filtered in the spleen. Lymph nodes contain aggregations of B and T cells along with macrophages and dendritic cells which interact with each other inducing adaptive immune responses.

Why Cardiovascular and **lymphatics are needed?**

Well, in unicellular organisms, nutrients can be absorbed directly from the environment in which a unicellular organism is living. However, a multicellular organism does not have this easy access to the nutrients. So, nutrients must surround the cells in a multicellular organism. These nutrients are provided by cardiovascular system which pumps blood to all organs. Remember that only fluid part of blood leaves blood vessels. RBCs never leave blood vessels. Sometimes proteins also leak from these blood vessels; however, once they leak from the blood vessels, they never can enter back into blood vessels because of their bigger size. If proteins remain in the tissue spaces, they exert osmotic pressure which attracts water into these space resulting in edema. However, nature has provided another mechanism in the form of lymphatic channels which start from tissue spaces and can drain tissue fluid and proteins back to the heart. This is the role of blood vessels and lympahtics that they ensure
circulation of nutrients throughout the body without causing any edema or abnormality. B cells and T cells also keep circulating between these two channels (cardiovascular and lymphatics). Unfortunately, microbes can also use these two systems to spread themselves in the body. That is the reason, spleen and lymph nodes are placed in these channels by the nature to deal with these microbes that find their ways into blood and lymph. Lymphatic System consists of the following: o Bone marrow o Lymph nodes o Spleen o Tonsils o Thymus o Mucosa associated o Bursa of Fabricius in birds

5. Disk diffusion method

A standardized inoculum of the organisms is spread on agar plate. Filter paper disks soaked with anti bacterials are then placed on the lawn of the organism. Plates are incubated at 37oC and results are recorded. Drugs diffuse from the paper disks radially and if the organisms are sensitive to the drug, a clear area with no growth around the disks develops. The size of the zone directly relates to the degree of sensitivity of the drug. This zone is compared to a standard table for recording the results of sensitivity test. Results are reported as sensitive or resistant or intermediate (between the two extremes).

6. How microbes develop resistance in antimicrobial?

7. Define immunity and difference between innate immunity and adaptive immunity. 10 marks

What is immunity?

Immunity: Ability of the body to protect itself from harmful effects of diseases.
Susceptibility: lack of immunity which leads to easy establishment of disease.

Types of immunity

Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).
Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.

First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.

Second line of defense: Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.

Adaptive Immunity: This will be studied in detail in a separate lecture soon.

Adaptive (Acquired): It is the third line of defense. Acquired by experience, specific to an organism.

Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.

Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response.
O Because of memory, body's immune response increase with every repeated exposure. It has a limit also!

Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.

Adaptive Immune Response can further be divided into:

Humoral immune response (antibody response by B cells or lymphocytes)
Cellular immune response (T cells are made which kills microbial harboring cells).

8. Agglutination and Neutralization 2 marks

Agglutination: Antibodies can agglutinate antigens which are then cleared as a group by macrophages
Neutralization: Toxins bind with antibodies and they become incapable of binding to their target receptors thus the toxins become harmless.

9. What is plaque 2 marks

Plaques are equivalent to bacterial colonies. Basically a plaque is an area where the virus infection has lysed the cells. These plaques are formed on cultured cells in a culture dish. A monolayer of the cells is infected with the virus to be enumerated and then the culture is overlaid with the semisolid agar so that infected and lysed cells do not move away from the site in the dish. Plaques can be counted easily much like bacterial colonies

10. Difference between capsule and slime

Difference between capsule and slime

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

Slime: If glycocalyx is loosely attached to the cell, it is called slime.

11. PCR method 3 marks

PCR goes through 3 step cycles for about 30 to 35 times. Each cycle consists of a denaturing stage (about 95oC), primer annealing (50 to 56oC) and extending stage (72oC). A primer is a fragment of DNA which is specific to the signature sequence. Two primers span the signature sequence actually. At the end of 30 or so such cycles, enough DNA accumulates that one can visualize on an agarose gel easily.

12. Type of T cell

T helper cells

TH1 (CD4+)

TH2 (CD4+)

T cytotoxic cells (CD8+)

13. Inhibits growth cell wall drug name

Penicillins and cephalosporins are the major antibiotics that **inhibit** bacterial **cell wall** synthesis. They are called beta-lactams because of the unusual 4-member ring that is common to all their members.

14. Microbiota factor and distribution composition

Nutrients: Availability of various nutrients determines the type of organisms present in a system. Similarly, physical and chemical factors also influence the type of microbiota. Other factors include pH, O2, and CO2. Host defenses, level of stress and age of an individual also matters for allowing microbiota of an individual.

15. Capsule function 2 marks

The capsule impairs phagocytosis and help the organisms evade the immune system; however, antibodies can help cells of the immune system in phagocytosis of such encapsulated organisms. Only a few bacteria produce capsule which include Strep pneumonia, Klbesiella pneumonia.

16. Viral replication stages

Viral multiplication goes through various stages in the cell. These include viral attachment, entry into the cell by pinocytosis or fusion with the plasma membrane, uncoating of the virus, biosynthesis of capsids and nucleic acids, maturation (assembly of the virus) and final release of the virus from the cell.

17. Economic benefits of fungi 3 marks

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively). However, if Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from Aspergillus niger; cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

18. Types of symbiosis 3 marks

Interaction of microbiota with the host leads to a relationship called symbiosis. There are a few outcomes of this symbiosis which we must know. The two participants of this relationship are microbiota and the host. At least one of these participants is benefited and sometimes both can be benefited from this mutual relationship. Different names or terms have been given to these mutual relationships as under:

Mutualism: both are benefited

Parasitism: one is harmed.

Commensalism: One is benefited, other is unharmed.

19. Inflammation and write its consequence 5 marks

Damage to body tissues triggers a local defensive response which is called inflammation. It causes redness, pain, swelling, heat, and loss of function of that organ

During inflammation, various mediators of inflammation are released, some of which include:

Histamine

Kinins: Almost all cells.
Prostaglandins by damaged cells
Leukotrienes by mast cells
Cytokines are released (TNF-alpha), and they cause release of acute phase proteins from the liver

20. Cause and diagnose of anaphylaxis

The patient also experiences difficulty in breathing. It is an emergency as it may kill the person. Anaphylaxis may be local or systemic.

- Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.

Hypersensitivity can be diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation. Moreover, IgE levels can be measured by ELISA. Measurement of IgE is more practical as nature of allergens is not known most of the time.

- Rx: Epinephrine & antihistaminic drugs provide a means of treatment.

21. Two name of bacteria which is capsulated

Bacillus anthracic, streptococcus pneumoniae and Klebsiella are examples of capsulated organisms.

22. What is Epidemiology and its types

The study of where and when diseases occur and how they are transmitted in a population.
There are three basic types of investigations relating to epidemiology

1. Descriptive Epidemiology
2. Analytical Epidemiology **3.** Experimental Epidemiology:

23. Target site of antibiotics

24. Capsomere

Capsid is made up of protein subunits called capsomeres that are arranged in 3 possible ways to give a helical, or icosahedral or complex symmetry to the viruses.
Capsomeres: Subunit called protomers aggregate to form capsomeres.

25. Character of antigen

Antigen: A foreign substance that provokes immunity is called an antigen.
More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably.
Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens.
Antigens are foreign to the body.
Degradability: Inert molecules are not antigenic.

26. Algae structure

Body of multicellular alga such as seaweed is called a thallus which consists of branched holdfasts (anchor alga to rock) stemlike hollow stipes and leaflike blades. There is no vascular tissue in these algae. Also, the stipe is not lignified or woody, so it does not provide support to the weed. Surrounding water provides the support for the thallus. Some algae have a gas filled body inside them which keeps them floating in the water. This gas filled structure is called pneumatocyst or float

27. Acid fast stanning 5 marks

Some bacteria such as Mycobacteria have a waxy material in their cell wall.
Specific name for this waxy substance is mycolic acid
Mycobacterium tuberculosis: Causative agent for tuberculosis

M. leprae: Causative agent for leprosy

- **Principle of Acid-fast staining**: Mycobacteria are lipophilic, not easy to stain

But once stained, they are resistant to acidic alcohol decolorization process.

- **Overall Procedure for Acid-fast staining**:

Carbol fuchsin: It is a primary stain Heat is used to enhance penetration of carbol fuchsin.
Acid alcohol is a decolorizer.
Methylene blue is a counterstain.
Please note, Mycobacteria are gram positive, however, they look red in acidfast staining.

28. Capsule staining 10 marks

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.
– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

– Capsulated organism make glistening colonies as seen in the accompanying diagram below:

Bacillus anthracic, streptococcus pneumoniae and Klebsiella are examples of capsulated organisms.
Thickness of the capsule depends upon the culture conditions.
Capsules are mostly are water soluble. Also remember that capsular organisms usually make the broth viscous and stringy
Capsule staining procedure

1. Prepare a thick smear in a loopful of congo red (1%) stain
2. Fixation in acid alcohol for 15 sec
3. Wash with dH2O
4. Cover it with acid fuchsin for 1 min.
5. Wash with water
6. A bacterium stains red, capsule remains colorless and is seen in the dark blue background as seen in the accompanying diagram.

29. Define vaccine and its types 10 marks

What is vaccine:

A product that creates immunity against an infection is called a vaccine. It could be the whole organism or a part of the organism.
The observation that people or animals once recover from an infection do not get sick easily to that organism again led to the discovery of vaccines and vaccination.

Types of vaccine:

Live Attenuated Vaccines: These are weakened organisms so that they do not harm the body, but divide or replicate in the same way as pathogenic version of the organism does in the body. So, when a live virus or bacteria multiplies in the body, it stimulates the immune system to create an adaptive immune response which is protective against the disease for which vaccine was given.
The benefits of live vaccines are a better immunity because live microbes mimic actual infection (although they do not cause harm to the body). Live vaccines induce both humoral and cellular immune responses together, so live vaccines are a better choice if available. Because live microbes multiply in the body, this replication results in the increase of antigenic amount needed for stimulation of the immune system. Also, this antigen remains for a long time in the body, continuous presence of antigen continuously keeps the immune system active for longer time resulting in long term immunity.

Inactivated Killed Vaccines: Sometimes a non-pathogenic version of the microbe is not available, and to make a vaccine in this situation, the organism is killed with formalin or phenol and then used as a killed inactivated vaccine. Because organisms in such a vaccine do not replicate in the body, frequent boosters are required to achieve a good level of immunity (antibodies). Also, remember that this is a killed vaccine, so classical cellular immunity is not created with killed vaccines.

Toxoids: Some diseases are caused by bacterial toxins, and not by bacteria. In such cases, toxin is modified (inactivated) and used as a vaccine. Such a vaccine is known as a toxoid. Tetanus toxoid is a good example.

Conjugated Vaccines: Sometimes special parts of organisms are required for generating protective immunity. However, those substances are poor antigens if used alone. In order to enhance immunity, such poor antigens are attached to proteins, and such vaccines are called conjugated vaccines. Haemophilus influenzae type b is a good example.
DNA vaccines: We now know that sometimes only certain proteins are required for generating a good immune response. In such cases, the genes responsible for those proteins can act as vaccines. DNA coding for such a protein is injected into

cells, and that DNA gets transcribed by the cells of the body into the required protein and that protein induces the antibody production (because that protein is foreign to the body). West Nile Virus is a good example of DNA vaccine

Subunit vaccines: Also called recombinant vaccines, such vaccines are composed of a portion of viral or bacterial proteins that can induce immunity. In other words, this is another version of DNA vaccines. In subunit vaccines, antigenic epitopes are included in the genome which is inserted into an expression vector that makes those subunit proteins. These subunit proteins are collected and used as vaccines to stimulate the immune system. Again, these act like a killed vaccine. There is another mechanism being explored in these days for recombinant vaccines. A subunit vaccine for Foot and Mouth disease virus is available.

Such genes are being incorporated in plants now that are acting like an expression vector. Animals can eat those plants and subunit vaccines can induce immune response through M cells of the intestine. Success is limited though so far.

30. Describe features of fungi 5 marks

These are simple eukaryotic cells. Some are unicellular, others are multicellular (thallus); however, they lack tissues such as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are mostly photoautotrophs; however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the face of the earth. Microscopic exam is needed to identify unicellular and filamentous algae. However, multicellular algae that are commonly known as seaweeds are macroscopic in nature and can be identified morphologically without the help of a microscope. Four groups of such algae include blue-green algae, green algae, brown algae and red algae. These algae are located in the sea at various locations and absorb light of various wavelengths, hence red algae are located r from the surface and can use blue light from the sun as blue light is of shorter wavelength and can penetrate deep in the sea. Please see the accompanying figure (above) for a better understanding of how wavelength relates with various algae. Also remember, blue-green algae need magnification in order to be correctly identified, although they are not microscopic.

31. Treatment of hypersensitivity ii 3 marks

This type of hypersensitivity can be diagnosed by history and symptoms. Glomerulonephritis and cellulitis is very common. Deposition of immune complexes in the kidneys can be diagnosed by presence of IgG in kidney cells by using anti-IgG antibodies.

32. What are antibiotics 3 marks

antibiotics are used to kill organisms in disease states, some antibiotics are not very effective for this purpose; however, these antibiotics could be used in food to prevent food spoilage. Nisin and natamycin prevent spoilage of cheese

33. Define virion and viroid 2 marks

A **virion** is an infectious particle composed of nucleic acid that is surrounded by a protein coat (capsid). Viruses are classified based on their nucleic acid and protein coat. Viruses have an extracellular and an intracellular state. Outside of a cell, in the extracellular state, a virus is called a virion.

VIROIDS: They are composed only of circular single stranded (ss) RNA.

34. Write entry point of microorganism 2 marks

Mucus membranes: These include linings of digestive tract, respiratory tract, urogenital system, and conjunctiva

Skin: Surface area wise, skin is the largest organ of the body. Unbroken skin is impermeable to organisms

Parenteral Route: Injury to the skin or mucous membrane leads to easy access for the organisms in the body. Any cut received on the skin for example becomes a portal of entry

35. Role of thin clonal selection 2 marks

According to this theory, B cells are generated in the bone marrow randomly without seeing the antigen. When such a B cell encounters its specific antigen in the body, it starts proliferating into a clone which ultimately becomes plasma cells and start secreting antibodies. A beautiful example to understand clonal selection theory is like going to a shoe store and select a shoe size that fits you very well. Note, that the shoe that you selected was made randomly without any regard to your foot size.

35. Antigen presenting cell 3 marks

These cells are able to present antigens to B cells for making antibodies by B cells.

There are three cells that fall into this category: Dendritic cells, macrophages, and B cells

36. How antibodies provide defense against microbes 5 marks

37. Reverse transcription 2 marks

A reverse transcriptase is an enzyme used to generate complementary DNA from an RNA template, a process termed reverse transcription

38. Bactericidal and bacteriostatic drug 2 marks

Bactericidal: Those antibiotics that kill and lyse the cells are called bactericidal.

Bacteriostatic: Such antibiotics stop the growth of the organisms, they do not kill or lyse the cells. Cells of the immune system then clear them from the body

39. Define plasmogamy karyogamy and meiosis 3 marks

1.Plasmogamy: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-). Both cells are haploid (one set of chromosomes)

2. **Karyogamy** The (+) and (-) nuclei fuse to form a diploid zygote nucleus.

3. **Meiosis**: The diploid nucleus gives rise to haploid nuclei (sexual spores).

40. Mechanism of hypersensitivity iv 5 marks

It is called as delayed type because its bad effect becomes visible 2-3 days after exposure to the antigen. This is the only hypersensitivity that involves cells rather than antibodies. Robert Koch discovered this when he injected tuberculin in patients. Contact dermatitis is the most common example of this kind of hypersensitivity. This is basically an interaction between an antigen, T cells and macrophages. Transplant rejection is another good example of delayed type IV hypersensitivity. Reaction to hair dyes is another example. Poison Ivy is another example in which contact with this plant initiate type IV hypersensitivity response. Chemicals from poison ivy interact with skin proteins and change them into foreign-looking protein molecules. Body's immune system mounts an immune response against these proteins resulting in a damaging type of intense inflammatory response

41. Cytokines 2 marks

Cytokines are released as a result of defense cell stimulation by PAMPs and TLRs interactions. Cytokines recruits more immune cells and also stimulate adaptive immune response

42. Note on infectious disease transmission with vector vehicle and contacts?

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another. o Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.

Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another.

43. Incidence and Prevalence 2 marks

Incidence: # of new cases in a given time period in a population

Prevalence: New + existing cases

44. PRIMARY AND SECONDARY IMMUNE RESPONSE 5 marks

When a person is exposed to an antigen (a microbe for example), the antigen is presented to B cells which start making antibodies. Initially, there are no detectable antibodies in the serum. However, after 3-5 days, IgM antibodies are secreted and then 7-10 days after, the same B cell starts making IgG. The level of IgG rises to a level and then it starts declining. This is a typical response of an antigen exposure and is called a primary immune response. It is interesting to note that a second exposure with the same antigen results in a heightened response in terms of very high levels of antibodies that are achieved in a very short time (within 2-7 days). This is called a secondary immune response. It is also called a memory or anamnestic response because memory cells are involved that ensure a quick response to the second exposure by the same antigen (to which the immune system has already been exposed). The accompanying figure illustrates this concept (next page). Blue line in the figure depicts a primary and a secondary response.

Adaptive Immune Response can be acquired naturally or artificially. Each of these categories can further be divided either into active or passive type.

Naturally Acquired Active Immunity: This type of adaptive immunity develops in a person who gets infected with a microbe but then recovers from that illness making antibodies.

Naturally Acquired Passive Immunity: A typical example of this type of immunity is transfer of antibodies from colostrum or milk of a mother to the offspring.
Artificially Acquired Active Immunity: When a person receives a vaccine against a microbe and the body makes antibodies in response to that vaccine, it is called artificial acquired active immunity because body makes these antibodies actively by itself.
Artificially Acquired Passive Immunity: When premade antibodies are given to a person in order to protect a person from an illness or a snake bite, this constitutes an artificially acquired passive immunity type. The following figure illustrates these four types.

45. Vegetative Hyphae and Aerial hyphae 2 marks

Vegetative Hyphae: Hyphae that are embedded in the medium and are used to obtain food are called vegetative hyphae.

Aerial or Reproductive Hyphae: The portion of hyphae that is concerned with reproduction. This portion is projected into the air. They bear reproductive spores.

46. Signature sequence and primer

signature sequences

There are many tests that involve nucleic acid sequence detection for the presence of microbes in a clinical sample. Please remember, as antibodies are specific to the organisms, DNA or RNA sequences must be uniquely specific for the microbe that we are going to detect. Such sequences that are specific to an organism are called **signature sequences**

Primer

A primer is a fragment of DNA which is specific to the signature sequence
Feature of yeast cells

47. Active and passive transport

active Movement

Movement against the concentration gradient

From low concentration to high concentration of a substance

Requires energy (ATP)

Substances that are actively transported into the cells undergo either no change or they are chemically modified to keep them there in the cell. Remember, active transport brings substances inside the cell against the gradient; hence molecules brought in can go out. That is why these molecules may be modified to keep them inside the cells. In other words, this modification avoids passive diffusion of those molecules from the cells outwards.

Passive Movement

Movement of substances with the concentration gradient. This means that substances will move from higher concentration of a substance to lower concentration. Gradient provides the force for movements of molecules and this happens with no energy expenditures.

What are the 2 benefits algae get from oceans 2 marks

Algae are important for aquatic life. They are at the bottom of the food chain because they fix CO_2 into organic molecules (Carbs) which could be used by other organisms.
They produce 80% of molecular oxygen on earth.
Algal blooms which result from fluctuation of temperature sometimes may cause problems for nature but that happens occasionally.
Algal blooms tell us about the status of pollution in the environment. Planktonic organisms (dead and decaying) are responsible for oil production. Petroleum and natural gas are the residues (hydrocarbons) of diatoms and other planktonic organisms.

Define lichens 2 marks

Lichens are a combination of green alga (or a cyanobacterium) and a fungus. Green alga or cyanobacterium is also called a phycobiont and the fungus is called a mycobiont in this relationship. Both partners have a symbiotic relationship with each other

Hypersensitivity type 1 3 marks

It involves IgE and it occurs within 30 minutes of exposure to the allergen. Antigens called allergens preferentially induce IgE production instead of IgG against the antigens. These IgEs get bound to mast cells in the tissues and a second exposure to the same allergen activates mast cells to secrete histamine and other mediators of inflammation. Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.
Hypersensitivity can be diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation

Transmission of disease 5 marks

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.
Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.

Droplet Transmission: Mucus droplets are created when you sneeze for example. These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

Hypersensitivity type3 5 marks

This hypersensitivity involves soluble antigen circulating in the serum and antibodies (IgG or sometimes IgM). Complement gets activated when these complexes get deposited in capillary beds as we see in the kidneys for example. A typical example will be the use of antivenom (antibodies) that was prepared in horses but given to a human to save the life. What happens is that horse antibodies will be recognized by human body as a foreign protein and human immune system will generate antibodies against this. This type of hypersensitivity can be diagnosed by history and symptoms. Glomerulonephritis and cellulitis is very common. Deposition of immune complexes in the kidneys can be diagnosed by presence of IgG in kidney cells by using anti-IgG antibodies.

BEST OF LUCK

1. Karyogamy plasmogamy and meiosis? 2MARKS (PPT 58)

Plasmogamy: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-) Both cells are haploid (one set of chromosomes).
Karyogamy: The (+) and (-) nuclei fuse to form a diploid zygote nucleus. **Meiosis:**
The diploid nucleus gives rise to haploid nuclei (sexual spores).

2. Advantage of Dry Heat relatively? 2 MARKS

Advantages of dry heat sterilization include: it is inexpensive, it does not pollute or cause toxic fumes, and it does not corrode or rust metal objects.
Disadvantages include: it is relatively slow and many objects cannot withstand the high temperatures.

3. How Microorganism cause damaged the body?

Infection with a **pathogen** does not necessarily **lead** to disease. Infection occurs when viruses, **bacteria**, or other **microbes** enter your **body** and begin to multiply. ... Pathogenic **microbes** challenge the immune system in many ways. Viruses make us sick by killing cells or disrupting cell function.

4. Microbial Identification?

Microbial identification can be **defined** as "**microbial** characterization by a limited spectrum of tests pre-chosen and appropriate to the problem being studied"

Methods: Classical or Conventional Method, Serological testing, Nucleic acid based testing.

5. Write two method? Viral Replication and Its mechanism? Stages of Viral Replication?

Viral replication involves six steps:

1) attachment,
2) penetration,
3) uncoating,

4) replication, **5)** assembly, **6)** release.
... During uncoating, replication, and assembly, the viral DNA or RNA incorporates itself into the host cell's genetic material and induces it to replicate the viral genome.

6. Immune and Its Types? Define Imunity and Its types? Innate and adaptive immunity? (10)

Immunity is the balanced state of multicellular organisms having adequate biological defenses to fight infection, disease, or other unwanted biological invasion, while having adequate tolerance to avoid allergy, and autoimmune diseases.

Innate Immunity

Plants and animals have what is called innate immunity. Innate immunity is the first line of defense against pathogens. It involves several cell types, proteins, and even an organ. The organ involved is your skin. Yes, skin is part of the first line of defense. It protects you and prevents pathogens from getting inside your body.

Adaptive immunity is an important part of the immune system. It is protection from an infectious disease agent that is mediated by B- and T- lymphocytes following exposure to specific antigen, and characterized by immunological memory.

7. Economies use of fungi?

They play an important **role** in medicine yielding antibiotics, in agriculture by maintaining the fertility of the soil and causing crop and fruit diseases, forming basis of many industries and as important means of food.

8. Gram Staining Procedure?

Crystal violet is added. – Primary stain • Crystal violet is washed off. • Iodine is added to enhance binding. – Mordant. • The slide is washed off with alcohol. – Decolorizing agent • Gram positive bacteria retain crystal violet • Gram negative bacteria appear colorless. • Crystal violet and iodine make a complex in the cytoplasm • Peptidoglycan layer is thicker in Gram positive bacteria and CV-I is retained. • Alcohol is rinsed off and stained with safranin. – Counterstain • The smear is washed again. • Blotted dry and examined microscopically.

9. Difference Simple and Differential Media?

Selective and differential media are used to isolate or identify particular organisms. Selective media allow certain types of organisms to grow, and inhibit the growth of other organisms. ... Differential media are used to differentiate closely related organisms or groups of organisms.

10. Horizontal and Vertical Gene Transfer? 3

Vertical gene transfer – Parents to offsprings

Horizontal – From cells to cells within the same generation – From bacteria to bacteria.

11. Difference LB Broth and LB Agar?

LB Broth: **broth** (**LB**) is a nutritionally rich **medium** primarily used for the growth of bacteria. Its creator, Giuseppe Bertani, intended **LB** to stand for lysogeny **broth**, but **LB** has also come to be commonly referred to as **Luria broth**, Lennox **broth**, or**Luria**-Bertani **medium**.

LB Agar:

Luria broth (**LB**) is a nutrient-rich media commonly used to culture bacteria in the lab. The addition of **agar** to **LB** results in the formation of a gel that bacteria can grow on, as they are unable to digest the **agar** but can gather nutrition from the **LB**within.

12. Chemical used in Gram staining?

Both gram-positive and gram-negative cells have peptidoglycan in their cell walls, so initially, all bacteria stain violet. Gram's iodine (iodine and potassium iodide) is applied as a mordant or fixative. Gram-positive cells form a crystal violet-iodine complex. Alcohol or acetone is used to decolorize the cells.

13. Viruses, Virion, and Virusoid and Prion?

Viruses: Acellular organisms – Proteins and nucleic acids

Viroids: composed only of circular ssRNA (potato spindle tuber viroid)

Virusoids: Contain circular ssRNA, need helper viruses for replication and encapsidation; also called satellite viruses

Prions: Infectious proteins

14. Uses of UV in Lab?

- Transilluminators. **Used** for the visualization of molecular samples, transilluminators are indispensable tools for researchers at life-sciences **laboratories**. ...
- **UV** Crosslinkers. ...
- Fluorescence Analysis Viewing Cabinets and Work Stations. ...
- **Ultraviolet** Lamps. ...
- Digital Radiometers/Photometers

15. Epidemiology and Its type?

Monitoring diseases in a population • Detection of outbreaks and developing epidemics • Implementation of appropriate control measures • Surveillance studies.

There are four most common types of epidemiological studies:

- Cohort Study.
- Case Control Study.
- Occupational Epidemiological Study.
- Cross-Sectional Study.
- Epidemiologists use statistical methods to determine whether the differences they see are real or due to chance fluctuations.

16. Streaking Plates method?

Streak plate technique is used for the isolation into pure culture of the organisms (mostly bacteria), from mixed population. The inoculum is streaked over the agar surface in such a way that it "thins out" the bacteria. Some individual bacterial cells are separated and well spaced from each other.

17. Process of bacterial identification?

The **identification** of **bacteria** is a careful and systematic **process** that uses many different techniques to narrow down the types of **bacteria** that are present in an unknown **bacterial** culture. It produces benefits for many aspects of the research of microorganisms and helps physicians correctly treat patients.

18. Physical Method of Sterlization?

Heat **Method of Sterilization**

This is the most common **method of sterilization**. The heat used kills the microbes in the substance. ... In heat**sterilization** process, the longer the exposure to heat the better is the **sterilization** at a given temperature.

19. Disadvantage of Chlorine Sterilization?

disadvantages as well; First of all high concentration of this agent corrodes metals and also damages cloths. Secondly the strength of this solution decreases with time so whenever you want to use it always prepare its fresh solution.

20. Advantage of dry heat Sterilization?

Advantages of dry heat **sterilization** include: it is inexpensive, it does not pollute or cause toxic fumes, and it does not corrode or rust metal objects.**Disadvantages** include: it is relatively slow and many objects cannot withstand the high temperatures.

21. Virulence

The extent of pathogenicity is called virulence of a pathogen.

22. Name the three subgroups of spiral bacteria

These are curved shaped bacteria. They are further divided into 3 more subgroups. a.

Vibrio: curved rods

b. Spirillum: Helical but rigid

c. Spirochete: Helical but flexible

Spirochetes move by axial filaments which are enclosed by the outer membrane

23. flagellum and function (2

Flagellum: Long filamentous structure that propels bacteria It helps the bacteria in movement.

24. Name the Types of Bacteria on the basis of Oxygen Requirement

a. Obligate Aerobes: Do not grow without oxygen. It is necessary.
b. Facultative Anaerobes: Oxygen is necessary but start become anaerobes in its absence and can grow without of O_2.
c. Obligate Anaerobes: Cannot grow in the presence of oxygen.
d. Aerotolerant aerobes: Do not need O_2, also do not bother its presence
e. Microaerophiles: Aerobes but need small amount of oxygen. Cannot grow if available in large amount.

25. conjugation, transformation and transduction (3)

Transformation

Naked DNA in solution is transferred from one bacterium to another during this process.

Conjugation

Genes transfer process can be mediated by plasmids. Plasmids are extra chromosomal circular DNA fragments that replicate independently of bacterial chromosome. Conjugation requires direct cell to cell contact and cells have to be opposite mating types.

Transduction

Transferring of a gene from a bacterium to another bacterium via a virus is called transduction. o When a bacteriophage infects a bacterium such as E. coli, it replicates inside E. coli and also produces a protein coat (which is called a capsid) in which viral DNA is packed before the virus is released.

26. how disease can be transmitted (5)

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.

Droplet Transmission: Mucus droplets are created when you sneeze for example. These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

What are Vectors? A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another.

Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another. Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another

27. Any two microbial identification method (5)

Classical or Conventional Method

Serological testing

Nucleic acid based testing … plz See detail on 49 to 54 lessons on handouts

28. epidemiology and methods (5)

The study of where and when diseases occur and how they are transmitted in a population

Descriptive Epidemiology: (recording the data about disease). It includes location and time of cases of disease, gender, health, age of patients etc. (Basically, frequency and distribution of risk factors in population are recorded). Prevalence and incidence of disease are typical examples. Such studies can be retrospective (after they occur) or prospective (before they occur). So, in summary, in descriptive epidemiology, information about an already existing disease is recorded.

Analytical Epidemiology: It relates to determining the cause of the disease. Analysis of data, mode of transmission of diseases and means to prevent diseases also come in this discipline. Such studies can be done with case control methods in which factors that have preceded the disease are determined. A group of people who have the disease is compared with another group of people in the same location who are free of the disease. A Cohort method tracks two groups forward from exposure to outcome. Cohort study compares the experience of a group exposed to the factor with another group not exposed to the factor. An association between exposure and its outcome is determined which makes it easy to see the risk factor involved. In other words, cohort studies begin with a group of people (a cohort) free of disease. The people in the cohort are grouped by whether or not they are exposed to a potential cause of disease. The whole cohort is followed over time to see if the development of new cases of the disease (or other outcome) differs between the groups with and without exposure.

Experimental Epidemiology: A hypothesis relating to a disease is tested in this approach. For example, smoking cause cancer can be tested by designing an experiment in mice in which they are exposed to certain levels of smoke over time and then the outcome is observed. Another example will be testing of a drug for prevention of a disease.

29. Magnetostome (5)

Inclusions of iron oxide

Surrounded by invaginations of plasma membrane

Present in G negative bacteria

Act like a magnet. Bacteria can stick to iron containing rocks for nutrition.

Decompose H_2O_2 which is toxic for cells.

30. Define immunity, difference between innate and adaptive immunity in Detail (10)

Immunity: Ability of the body to protect itself from harmful effects of diseases whereas Susceptibility is the lack of immunity which provides the pathogens easy establishment to cause the diseases Types of Immunity Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens acts in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).

Does Not Have Memory. Do not increase in responses to the repeated exposure to the same agent. Therefore, it has no memory.

First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.

Second line of defense: Phagocytes and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaking the first barrier, macrophages, neutro-phils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.

Adaptive (Acquired): It is the third line of defense. It is acquired by experience and specific to an organism.

Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.
Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response.

Increases with repeated exposure: Because of memory, body's immune response increase with every repeated exposure. It has a limit also!

Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.

Adaptive Immune Response can further be divided into:

Humoral immune response (antibody response by B cells or lymphocytes) and Cellular immune response (T cells are made which kills microbial harboring cells)

31. cell membrane function

a. Selective permeable barrier
b. Passive and Active Transport
c. Respiration in microbes (bacteria)
d. Photosynthesis in microbes
e. Lipid synthesis
f. Cell wall parts are transported by a molecule called bactoprenol which is present in the plasma membrane.
g. PM has many receptors in it with which various ligands can bind and initiate signals for gene expression etc.

32. Glycoclyx, its composition and function

It is the outermost layer of the cell wall, viscous and gelatinous in nature that surrounds the cells. It is composed of polysaccharide and polypeptide or both. If glycocalyx is organized and firmly attached to the cell, it is called a capsule. – Plays important role in virulence (degree of pathogenicity) – Can also be a part of the vaccine against the bacteria to which it belongs.

Glycocalyx is also part of biofilms that bacteria make to attach to surfaces. Bacteria secrete extracellular polymeric substance (EPS) Biofilm protects cells within it.

It also facilitates communication amongst cells

If glycocalyx is loosely attached to the cell, it is called slime.

33. Differentiate between vertical and horizontal gene transfer and name three horizontal gene transfers

Vertical Gene Transfer:
This is a normal way of transferring genes from parents to offspring. This happens when a cell divides. Each daughter cell receives exactly what its parent cell has.

Horizontal Gene Transfer:
When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer. This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:

1. Transformation
2. Conjugation
3. Transduction

34. three types of passive movement

Diffusion Facilitated Difussion Osmosis

35. Economic benefits of fungi

Saccharomyces are used for alcoholic beverages under anaerobic conditions and if incubated aerobically, it is used in bread making because it metabolizes glucose to produce CO_2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise.

This yeast is also extensively used for molecular biology work and vaccine production.

Vitamin C is obtained from Aspergillus niger.

Cellulase from Trichoderma
Antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal

Entomophaga (kills gypsy moth), hence is used as a biological pest control.

Mushrooms can be cooked and eaten as a source of proteins for humans.

36. Pili and its functions

These are hair-like structure composed of pilin, usually one to ten in number. Longer than fimbriae Used for attachment to:

Host cells

Bacteria

Used for DNA transfer from one bacterium to another during Conjugation (Sex pili)

Also function in twitching Motility

Gliding Motility is also the function of the pili.

37. physical nature of media 2

Solid Media

Liquid Media

38. germination of spore, 5

Exosporium: A thin delicate outermost covering of the spore

Spore coat: 2 nd layer underneath the exosporium. It is thick and composed of several protein layers. Resistant to

It contains enzymes for germination. Germination of spores into vegetative form occurs when environment becomes favorable for their growth. Cortex: It is the 3rd layer from outside in. It has peptidoglycan in it.

Spore cell wall or core wall: Surrounds the protoplast or spore core Spore core: Contains nucleoid and

chemicals

ribosomes

39. Give two advantages of solid media in bactrial culturing.2

Solidification of the medium is done for purifying organisms from each other as solid medium provides surface for individual colonies to grow well separated from each other.

Secondly, the solid media are used to study the colony characteristics.

40. Types of microscopy 5

- Dark field microscopy
- Phase contrast microscopy
- Fluorescent microscopy
- Transmission electron microscopy
- Scanning electron microscope

41. Writes stages of respiration of glucose 3

Glycolysis: Oxidation of glucose to pyruvic acid Krebs

cycle: Oxidation of acetyl CoA to CO2

Electron Transport Chain: Coenzymes that carry electrons from Krebs cycle or glycolysis are oxidized to create ATP.

42. Differentiate between horizontal and vertical gene transfer.

- **Vertical Gene Transfer** o This is a normal way of transferring genes from parents to offspring. This happens when a cell divides. Each daughter cell receives exactly what its parent cell has.

- **Horizontal Gene Transfer**

genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer.

 When

This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:

1. Transformation 2. Conjugation 3. Transduction

43. Define buffers.

- Buffers are used to maintain desired pH.
- Bacteria grown in lab produce acids. o Growth inhibited by acids. o Peptones and amino acids are used as buffers. o Phosphates are also used for buffers.

44. Define immunity, innate immunity and adaptive immunity.

- **Immunity: Ability of the body to protect itself from harmful effects of diseases.**
- **Innate (Natural):**

- Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific. o Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body. o Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there). o Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.

- **First line of defense:** Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside. o **Second line of defense:** Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body. **Adaptive Immunity:**

The **adaptive immune** system, also known as the acquired **immune** system or, more rarely, as the specific **immune** system, is a subsystem of the overall **immune**system that is composed of highly specialized, systemic cells and processes that eliminate pathogens or prevent their growth.

Adaptive (Acquired): It is the third line of defense. Acquired by experience, specific to an organism.

45. What is epidemiology? Describe its tyes.

- The study of where and when diseases occur and how they are transmitted in a population. There are three basic types of investigations relating to epidemiology:

Descriptive Epidemiology

Analytical Epidemiology Experimental Epidemiology:

46. Enlist microbial identification methods.

Classical or Conventional Method Serological

Methods

Nucleic Acid based Methods

47. Write economical uses of fungi.

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively). However, if Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO_2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from *Aspergillus niger*; cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

48. What is sporulation?

• Formation of spores takes place within a vegetative cell and the process is called sporulation or sporogenesis. It is initiated when nutrients become unavailable.

49. Differentiate between viable and direct count.

50. What is flagellum? Write its functions.

Flagellum: Long filamentous structure that propels bacteria

51. What is meant by gene transfer? Name the types.

Vertical Gene Transfer
Horizontal Gene Transfer

1. Transformation 2. Conjugation 3. Transduction

52. Describe classification system proposed by Robert H Whittaker.

In 1969, five kingdom classification was proposed by Robert Whittaker as under:

•Plantae: plants

•Animalia: Animals

•Fungi: Yeasts, molds and mushrooms

•Protista: These are unicellular eukaryotes. Organisms that do not fit into any other category are placed in Protista. They are larger than prokaryotes. They include algae, protozoa, slime molds and water molds.

•Prokaryotae or Monera: Bacteria

53. What is taxonomic hierarchy?

All organisms can be grouped into a series of subdivisions that make up the taxonomic hierarchy.

Specie:

A bacterial **species** represents —a monophyletic and genomically coherent cluster of individual organisms that show a high degree of overall similarity with respect to many independent characteristics, and is diagnosable by a discriminative phenotypic property‖

Genus

Order

Family

Group

Class

Phylum

Kingdom

Domain

54. What is western blotting?

Western Blotting: We can use this technique for the detection of antigen in the serum. Microbial proteins can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) is present on the gel. Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies

55. Enlist types of passive transport.

Simple Diffusion

Facilitated Diffusion
Osmosis

56. Name types of spiral bacteria.

These are curved shaped bacteria. They are further divided into 3 more subgroups.

Vibrio: curved rods

Spirillum: Helical but **rigid**
Spirochete: Helical but **flexible**

Spirochetes move by axial filaments which are enclosed by the outer membrane.

57. Define passive movement.

Passive Movement

Movement of substances with the concentration gradient. This means that substances will move from higher concentration of a substance to lower concentration. Gradient provides the force for movements of molecules and this happens with no energy expenditures.

58. What is function of lysozymes?

A lysosome is a membrane-bound organelle found in nearly all animal cells. They are spherical vesicles that contain hydrolytic enzymes that can break down many kinds of biomolecules. Lysosomes are organelles that contain digestive enzymes. They digest excess or worn out organelles, food particles, and engulfed viruses or bacteria.

59. What are fimbriae? Give location and function.

- Hair-like small appendages on G-neg cells Composed of pilin, a protein Can be at one pole or around the entire cell Used for attachment to surfaces or epithelial cells Fimbria: Singular

Fimbriae help bacteria attach to surfaces

before they can secrete biofilm; E.coli O157: Diahrrea

• Fimbriae also help attach to epithelial cells e.g *Neisseria gonorrhoeae*

60. Name types of culture medium on the basis of function.

a. Differential medium
b. Selective medium
c. Enriched medium
d. Basal medium
e. Reducing medium
f. Selective medium
g. General purpose medium

61. Write functions of cell wall, flagella and ribosomes.

Cell wall gives support and protection against plasmolysis.

Flagella help bacteria in movement.

Ribosomes are organelles that assist in protein synthesis during translation.

62. What are types of microscopes?

a. Light microscope
b. Electron microscope
c. Scanning electron microscope
d. Transmission electron microscope
e. Dark field microscope
f. Bright field microscope
g. Fluorescent microscope

63. What do you know about types of techoic acid?

G-positive cell wall contains teichoic acid which may be divided into wall teichoic (remains within the cell wall) acid and lipoteichoic acid (is inserted into the cell membrane).

Teichoic acid is antigenic in nature and helps the cell wall by providing rigidity.

64. Write characteristics of integral proteins.

i. Integral proteins act as channels or carriers in facilitated diffusion. Integral proteins are called transporters or permeases ii. No energy is required during transportation of substances through these integral proteins.

iii. Two kinds of transporters are known: i. Nonspecific transporters
1. Ions ii. specific transporters

1. Change in shape 2. Sugars, vitamins

65. What are benefits of solid medium?

Solidification of the medium is done for purifying organisms from each other as solid medium provides surface for individual colonies to grow well separated from each other. A well isolated colony is assumed to be the progeny of a single cell. Secondly, the solid media are used to study the colony characteristics.

66. Define refractive index.

Refractive Index: Ability of a medium to bend the light o Light rays move in a straight line through a single medium o Light rays bend when the pass from one medium to another

67. Define medical microbiology.

Medical microbiology is a branch of **medical** science concerned with the prevention, diagnosis and treatment of infectious diseases. In addition, this field of science studies various **clinical** applications of microbes for the improvement of health.

68. What is resolution?

Resolution is the ability of the lenses to distinguish between two closely lying objects as separate.

$$d = \frac{0.5\lambda}{n \sin \theta}$$

o Light microscope resolving power is 0.2 µm.

69. Write three differences between prokaryotes and eukaryotes.

Prokaryotes

• DNA is not enclosed in a nuclear membrane.
• Chromosome: mostly Single, circular
• DNA not associated with histones
• No membrane enclosed organelles
• Cell wall has peptidoglycan, a complex
carbohydrate.
• Divide by binary fission.
Eukaryotes

• DNA is enclosed in a membrane-bound nucleus.
• DNA is found in multiple chromosomes.
• Chromosomes are linear, thread-like structures.
• DNA is associated with histones.
• Cell wall, if present, is chemically simple. Cellulose or Chitin

• They have membrane enclosed organelles.
Mitochondria chloroplasts Endoplasmic reticulum Golgi apparatus Lysosomes etc

70. Define virulence. Virulence: The extent of pathogenicity

71. Write methods for

1. Plate Counts

a. Pour Plate Method:
b. Spread Plate Method:
2. Filtration:
3. The most Probable Number Method:
4. Direct Microscopic Count

72. Write steps of germination of spores.

• Spore germinates when it finds a conducive environment for its growth. Germination has 3 phases or stages:

Activation: It prepares the spore for germination. onment.

Germination: Spore starts swelling and loosing its coats etc.

Outgrowth: New components are made.

- Heat can activate the spore when appropriate moisture and nutrients are present in the envir
- It becomes metabolically active.

73. Write types of lipopolysaccharides. 2

74. Define culture, media and inoculum.

Microbial growth in the lab is called a **culture**.

A nutrient material that supports the growth of microbes in the lab is called a **culture medium**.

Microbes introduced into a culture medium that initiate growth of organisms are known as **inoculum.**

75. Define antiseptics.

Compounds that are used to remove pathogens from living tissue are known as antiseptics.

76. Define phosphorylation.

The addition of an inorganic phosphate group to a chemical compound is called phosphorylation. e.g conversion of ADP into ATP in living tissues.

77. Explain phases of growth curve.

In a batch culture, organisms go through various phases of growth. If we follow this growth with respect to time, and make a graph of this growth pattern, it will give us what we call as a growth curve. There are four distinct phases of this curve. **1-Lag phase 2- Log phase 3- stationary phase 4- Death phase.**

78. What is high frequency of combination?

In some cells that carry F_+, F_+ (Fertility factor) gets incorporated in the chromosomal DNA which converts F_+ cells to high frequency of recombination cells (Hfr cells). When conjugation occurs between an Hfr cell and an F_- cell, the Hfr cell's chromosome (with its integrated F factor) replicates, and a parental strand of the chromosome is transferred to the recipient cell. Since this transfer starts from the middle of F factor gene, and most of the time, this transfer is not complete; the whole chromosome of Hfr cell is not transferred. However, some genes can be transferred during this process. When these genes become integrated into the genomic DNA of the recipient cells, the recipient cells acquires new versions of genes that were not part of its genome previously. It may be noted that F_- cells remains F_- negative cell because F factor is not transferred completely. This whole process is illustrated in the accompanying diagram.These newly acquired genes can be mapped easily with respect to the time they get transferred to F_- cells.

79. Differentiate between clone and strain?

Clone: Population of cells derived from a single cell that are genetically identical.

Strain: A genetic variant of a clone is called a strain.

80. Differentiate between positive sense and negative sense RNA.

81. What is zoonosis?

Zoonosis: Primarily an animal infection, but can be transmitted to humans. It can assume epidemic proportions in humans and is difficult to control or eradicate.

82. What are interferons? Give their function.

Interferons are antiviral proteins, produced by macrophages and lymphocytes in response to viral infections. Fibroblasts can also secrete it. These proteins interfere with viral replication. Thereare two types of interferons:

Species specific: This means that interferon produced by humans cannot be used in dogs or cats or vice versa.

Not viral specific: This means that interferons produced in response to virus A can also be effective against virus B.

Long Questions 5 & 10 marks.

83. Explain characteristics of aerophiles.

84. Describe conjugation, transduction and transformation. Horizontal gene 10

85. transfer:

When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer. This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:

1. Transformation 2. Conjugation 3. Transduction **1. Transformation** o Naked DNA in solution is transferred from one bacterium to another during this process. Frederick Griffith observed this process in 1928 when he was working with *Strep pneumonia* in his lab. This bacterium has two strains: one is capsulated and is called a Smooth strain (which is pathogenic) and the other is Rough strain (non-pathogenic).
o When live encapsulated bacteria are injected into a mouse, the mouse gets killed by the microbe. When rough strain is injected, the mouse is not killed. Moreover, when heat-killed encapsulated bacteria are injected, it also does not kill the mouse. However, when heat-killed strain is mixed with rough live strain, and a mouse is injected with this mixture, the mouse dies. If you isolate the bacterium from these dead mice, the bacteria resemble with encapsulated strain of *Strep pneumonia*. The interpretation of this experiment led to the mechanism that rough live bacteria take up DNA from the heat-killed encapsulated strain and becomes pathogenic by acquiring the ability to form a capsule. Very few bacteria have this property of accepting foreign DNA and most of it is degraded (only small fragments of DNA get incorporated in the genome of the recipient. The cells that can take up DNA are called competent cells. We can create such microbes in the lab that can accept a given DNA during gene transfers that we undertake in our DNA work. The cells that take up the DNA and incorporate this in their genome successfully are called transformed cells.

2. **Conjugation** o Genes transfer process can be mediated by plasmids. Plasmids are extra- chromosomal circular DNA fragments that replicate independently of bacterial chromosome. In other words, a given bacterial cell can have multiple copies of plasmid with the same cell. Plasmids often have genes that are not strictly needed by the microbe for its growth. However, the genes on the plasmids provide some advantages to the bacterium. For example, antibiotic resistant genes offer a mechanism for the bacterium to resist the
presence of that antibiotic. In other words, bacterium that has a plasmid with an antibiotic gene on it can still multiply in the presence of that antibiotic.

- Conjugation requires direct cell to cell contact and cells have to be opposite mating types. Plasmids in G –ive cells have genes that code for sex pili because pili are needed to make a bridge between the two mating type cells. Since there is no pilus present in G +ve bacteria, they secrete sticky surface molecules
instead which create contact between the two cells. The pilus along with two mating cells is shown in the diagram below:
o These plasmids are transmissible from one bacterium to another during conjugation.

- Plasmid is replicated and a single copy of the plasmid is transferred to the recipient bacterium through the pilus.

- A complementary copy is then made in both the cells to make the DNA of the plasmid double stranded molecule.
  - In E. coli, fertility factor (F factor) plasmid was the first plasmid observed to be transferred from one organism to the other, hence those bacteria that have this plasmid are called F_+ cells to differentiate from those that do not have one (F). However, once F_- cells acquire F_+ plasmid, they also become F_+.

3. Transduction o Transferring of a gene from a bacterium to another bacterium via a virus is called transduction. o When a bacteriophage infects a bacterium such as E. coli, it replicates inside E. coli and also produces a protein coat (which is called a capsid) in which viral DNA is packed before the virus is released. What happens is that when phages cause lysis of infected cells, bacterial chromosomal DNA also gets fragmented and such fragments can sometimes be packed into the viral capsids much like the viral DNA. Now, the capsid carrying the bacterial gene or genes in that fragment can infect another E. coli and can inject its DNA (bacterial DNA fragment) into it resulting in the transfer of genes from one bacterium to another.

85. Explain methods of microbial control in detail.

These methods can be broadly classified into four groups: Physical, Mechanical, Chemical and Biological

Physical methods of microbial control

Heat: It is the cheapest sources of all and easily available to control microbial growth. There are a few heat related concepts that we must appreciate. Heat denatures enzymes.

Thermal Death Point: It is the lowest temperature at which all cells in a culture are killed in 10 min.

It will be a specific temperature for a specific species of organism.

Thermal Death Time: Time (minimum) during which all cells in a culture are killed at a given temperature. This will vary from temperature to temperature for the same organism. Obviously, higher temperatures will take less time to kill the organisms than low temperatures.

rther be classified into **dry and moist** heat methods.

Heat kills by oxidation

Form of dry heat include:

1. Dry heat (hot-air oven),
2. Flaming (Platinum loop sterilization in the flame of Bunsen burner),
3. Incineration (burning to ashes).

$_{o}$C, and Tyndallization (100$_{o}$C for 3 consecutive days to sterilize sugar solution that can be degraded by autoclaving) and pasteurization (usually at low temperature than 100$_{o}$C).

Pasteurization: This technique typically employs low heat for killing pathogenic and food spoiling bacteria in milk. However, it does not kill thermoduric organisms. Thermoduric organisms are not pathogenic though. Products other than milk, such as ice cream, yogurt, and beer, all have their own pasteurization times and temperatures, which often differ considerably. Dairy industry tests phosphatase to determine if pasteurization of dairy product has occurred effectively because phosphatase present in raw milk gets inactivated with effective pasteurization temperatures. The following three equivalent heat treatments can be given to achieve pasteurization of milk:

- 63°C for 30 min
- High-temperature short-time: 72°C for 15 sec
- Ultra-high-temperature: 140°C for 4 sec

Other Physical Methods

Low temperature inhibits microbial growth by slowing down chemical reactions.

Refrigeration: Materials stored at 2-8°C can last for a day or two.

Deep-freezing: Long term storage of materials is possible at minus temperatures in the deep freezers.

Lyophilization: Freeze drying is another method for long term storage of food and other materials.

High pressure deshapes and denatures proteins: This technique is used for preserving juices and wines.
Desiccation prevents metabolism: This method is very old and still in use today. Food stored in high concentrations of salt or sugar solution can keep for long.
Osmotic pressure causes plasmolysis. High salt concentrations for example can take the water content of the cells out leaving them starved of water.

Filtration: It is also a physical method. It can be used to filter air (using high efficiency particulate air filter) or liquid medium using membrane filters.

Radiation: o Ionizing and non-ionizing radiation o Ionizing Radiation: High energy waves that destroys microbes.

- Ionization radiations include gamma rays, X rays, or high-energy electron beams. The ionizing radiation possesses a wavelength shorter than that of nonionizing radiation, less than about 1 nm. All these short wavelengths cause ionization of water generating high reactive hydroxyl radicals which are damaging to the cells as they bind to DNA and results in mutations. Medical plastic supplies, medicines and meat products can be sterilized by radiation.

Non-ionizing Radiation: wavelengths longer than 1nm fall into non-ionizing radiation. The best example is UV light. UV light causes mutations in the DNA by dimerization of thymidine bases. These *thymine dimers* inhibit correct replication of the DNA during reproduction of the cell. The most useful wavelength is 260 nm. This wavelength is specifically absorbed by DNA. UV radiation is used to control organisms in the air mostly. However, surfaces are also sterilized by UV radiation. UV light cannot penetrate deep into the fluids, so only surface bacteria are killed.

ETHYLENE OXIDE FOR MICROBIAL CONTROL

o **Ethylene Oxide:** Many heat-sensitive items such as disposable plastic, petri dishes and syringes, heart-lung machine components, sutures, and catheters are sterilized by ethylene oxide. Ethylene oxide is a strong alkylating agent that kills by reacting with functional groups of DNA and proteins to block replication and enzymatic activity. It rapidly penetrates packing materials, even plastic wraps. EtO is explosive, supplied in a 10% to 20% concentration mixed with either CO_2 or dichlorodifluoromethane. A clean object can be sterilized if treated for 5 to 8 hours at 38°C or 3 to 4 hours at 54°C when the relative humidity is maintained at 40 to 50% and the EtO concentration at 700 mg/L.

CHEMICAL METHODS FOR MICROBIAL CONTROL

Chemicals are used to control microbial growth. Factors that influence the efficacy of a chemical disinfectant include the concentration of the disinfectant, presence of organic matter in the environment (matrix) where the disinfectant is being used, pH of the environment (disinfectants are more effective at acidic pH), and time of exposure to the disinfectant.

86. How infectious diseases are transmitted? Explain in detail.

1) how infectious disease can be transmitted explain it briefly transformission occur to contact ,vehicle n vector.(10)

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.

Droplet Transmission: Mucus droplets are created when you sneeze for example.

These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

Vectors o A **vector** is a vehicle (animate or inanimate) that carries a pathogen from one host to another. o Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.
Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another.

Nosocomial Infections o Hospital acquired infections are called nosocomial infections.

87. Explain salient features of viruses, virions and viroids.

VIROIDS

These are infectious agents that consist only of RNA and found in plants only. No capsids, no envelopes, just RNA. They are covalently closed circular ssRNAs about 250 nucleotides long. The circular RNA normally exists as a rodlike shape due to intrastrand base pairing, which forms doublestranded regions with single-stranded loops. They do not encode any gene products, so they are replicated by cellular DNA dependent RNA polymerase. A plant infected with such viroids

may not show any signs of illness. However, in other plants, they may cause severe disease. They cause the disease by RNA silencing process.

VIRUSOIDS o They are similar to viroids; however, they encode one or two gene products and need a helper virus to infect the host cells. The helper virus supplies gene products and other materials needed by the virusoid for completion of its replication cycle. Hepatitis D virusoid is a typical example which uses Hepatitis B virus as a helper.

PRIONS

o These are proteinaceous infectious particles that cause neurodegenerative diseases such as scrapie in sheep, bovine spongiform encephlopthy, and Kuru in humans. These are abnormal form of cellular proteins. How these proteins accumulate in the cell is not clear, although, some genetic components are known to be involved.

88. Explain methods of microbial identifications.

89. How fungi can be identified by biochemical testing?

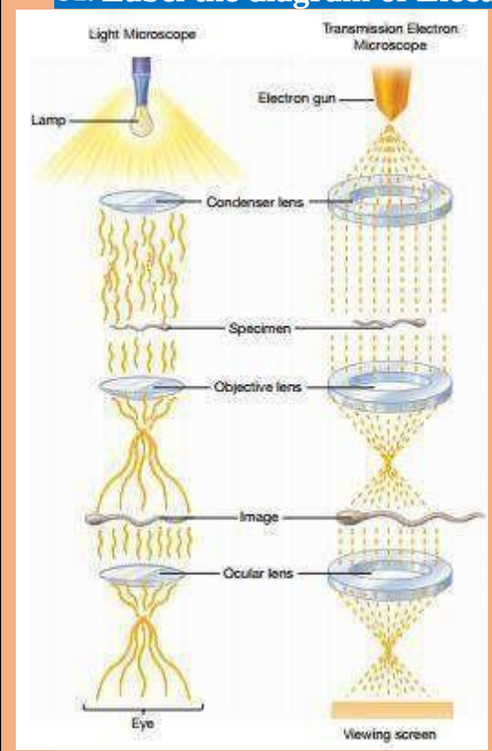
Yeast if grown on solid medium grows like bacteria. In other words, colonies of yeast look like bacterial colonies but are bigger in size.

 Yeasts can grow as facultative anaerobes. Saccharomyces if given oxygen to grow, it converts carbohydrates aerobically and produces CO_2 and water; however, if oxygen is denied, carbohydrates get converted fermentatively into ethanol and CO_2.

90. IDENTIFICATION OF SELECTED FUNGI

 Filamentous fungi are mainly identified by morphology, and not by biochemical testing. Fung slide (slide culture) on top of Sabouraud's agar. Spores are placed on top of the medium or embedded in the medium. A block of the medium can be used to grow fungi or a glass coverslip can be used to grow spores into hyphae and mycelium and spore arrangements can be studied under the microscope by staining with lactophenol cotton blue. Slides stained with lactophenol cotton blue can also be seen under the microscope.Stained slides can show specific arrangements of spores, their sizes and other morphological features as seen in the figure for Aspergillus.

91. Label the diagram of Electron microscope with numbers mentioned.



92. Write functions of cell membrane of bacteria.

- Selective permeable barrier
- Passive and Active Transport
- Respiration in microbes (bacteria)
- Photosynthesis in microbes
- Lipid synthesis

i are cultured

using a

• Cell wall parts are transported by a molecule called bactoprenol which is present in the plasma membrane.

• PM has many receptors in it with which various ligands can bind and initiate signals for gene expression etc.

93. Write names of 5 types of bacteria with respect to temperature.

Organisms are basically classified into three groups based on the temp requirements:

1. **Psychrophiles:** These are further divided into strict psychrophiles and psychrotrophs:

○ **Psychrotrophs:** Cold loving: 15 $_0$C ○ **Psychrotrophs:** Optimum temp is 20-30 $_0$C. Food spoilage bacteria that can spoil food during refrigeration.

2. **Mesophiles:** 25 – 40 $_0$C: These are the ones that cause diseases in animals and humans.

o Moderate temp loving organisms o Optimum: 37 $_0$C as this is the body temperature of humans and animals.

3. **Thermophiles:** 50 – 60 $_0$C

○ Heat loving o Important in organic compost piles o These are further divided into hyperthermophiles that grow optimally at 80 $_0$C. They live in hot springs. Every group of these organisms has a: ○ Minimum growth temp o Optimum growth temp. o Maximum growth temp.

94. Write characteristics of metachromatic granules.

Metachromatic Granules

• Also called volutin, they stain red with certain dyes such as methylene blue. That is why they are called metachromatic (stain in different color as methylene blue gives blue color but the color on these granules is red). • Large inclusions • These inclusions contain inorganic phosphates. Inorganic phosphates are used up in ATP synthesis. • Characteristics of *Corynebacterium diphtheria: This bacterium can be identified by the presence of these granules in it.*

95. Explain five kingdom classification systems.

In 1969, five kingdom classification was proposed by Robert Whittaker as under:

•Plantae: plants

•Animalia: Animals

•Fungi: Yeasts, molds and mushrooms

•Protista: These are unicellular eukaryotes. Organisms that do not fit into any other category are placed in Protista. They are larger than prokaryotes. They include algae, protozoa, slime molds and water molds. Prokaryotae or Monera: Bacteria and Archea

96. Explain layers of endospore.

Endospore and its various parts/structures

• **Exosporium:** A thin delicate outermost covering of the spore

• **Spore coat:** 2nd layer underneath the exosporium. It is thick and composed of several protein layers.

Resistant to chemicals

It contains enzymes for germination. Germination of spores into vegetative form occurs when environment becomes favorable for their growth.

• **Cortex:** It is the 3rd layer from outside in. It has peptidoglycan in it.
• **Spore cell wall or core wall:** Surrounds the protoplast or spore core

• **Spore core:** Contains nucleoid and ribosomes

97. Enlist types of bacteria on the basis of oxygen.

Based on the use and concentration of Oxygen, microbes can be classified into five groups:

Obligate Aerobes: Oxygen must be present for their growth. These organisms do not grow if O_2 is not present in the medium. If you test these organisms in a tube containing nutrient agar, the growth will remain confined to the surface only (see the diagram for the concept). Mycobacterium tuberculosis is an example of such an organism.
Facultative Anaerobes: These organisms prefer aerobic respiration; however, if oxygen is not available, they can use anaerobic respiration or fermentative modes for generating ATPs. These organisms will be present throughout the tube, however, more growth will be seen on the top (close to the surface) because these organisms prefer using oxygen if it is available.

Obligate Anaerobes: Although, these organisms do not use oxygen, they cannot grow in the presence of oxygen. The reason is that oxygen creates toxic compounds which can kill these cells. Normally, oxygen toxic compounds are eliminated by the cells, however, strict anaerobe do not have a system to get rid of these toxic compounds when they are made in the cells. So using or not using oxygen is one thing and growing in the presence of oxygen (and not utilizing it) is another. The growth will remain confined to the bottom which has little oxygen.
o **Aerotolerant Aerobes:** These organisms do not use oxygen and also are not bothered by the presence of oxygen. In other words, they have a better system to dispose of toxic oxygen compounds. They will grow evenly throughout the tube of medium

98. Name five types of microscopy . 5 marks

Dark field microscopy

- This microscope makes use of a special condenser that provides illumination from the sides only as depicted in the diagram below:
- This special arrangement of the condenser throws light rays at an oblique angle so that if the specimen is not present on the slide, there will be no image formed. However, if the the organisms are present on the glass slide, those organisms will reflect the light into the objective lens and we will see the cells as bright objects in a dark background much like we see stars in the sky at night.
- The beauty of this microscope is that one does not have to stain the organisms to see them under the microscope.

Phase contrast microscopy

This instrument is built on the concept of phases of light waves. The light ray that passes through an organism gets retarded about ¼ of its wavelengths than the light that passes outside the organism (through the glass slide only). To further retard the retarded light ray, a phase plate is inserted into the body of the microscope (above the objective lens) which is thicker in the center than the ends. This arrangement causes the retarded light to further slow down about ¼ of its wavelength, which when recombines with the un-retarded light creates a contrast which makes the cells/organisms visible.

- This method for visualizing organisms also does not involve staining.

Fluorescent microscopy o This microscope is equipped with UV light that provides the source of light for illumination. However, this light source does not pass through the slide, it instead falls over the slide (having specimen) and creates fluorescence.

- Fluorochromes are substances that absorb short wavelength of light and emit longer wavelength (visible light spectrum)

- Fluorochromes are attached to antibodies that are specific to organisms. So, if an organism is present in the specimen, we can use specific antibodies to determine the presence of those organisms by use of fluorochrome tagged antibodies. See the accompanying diagrams for details. Look for UV lamp (Mercury Lamp).

Transmission electron microscope

- Electron microscope (EM) makes use of electron beam as an alternative to light rays. Since electron beam has much shorter wavelength, resolution of EM is 0.1nm compared with light microscope which is 200nm.
- Second difference of EM with light microscope is that focusing of electron beam is done not by glass lenses but by electromagnets that acts like lenses.
- The third difference in EM is that the image of the specimen is formed on a fluorescent screen like a TV screen.

Scanning electron microscope

This is a modified version of electron microscope. Electron beam is allowed to fall on to the specimen and the image is formed from the reflected electrons that are collected and assembled into an image by the detector. It gives a three dimensional picture.

99. Explain ELISA.

There are some interactions (Antigens and Antibodies) that are not directly evident. Interactions may have taken place, but there is no visible direct clue if the interaction has taken place or not. For such interactions, we use indirect methods. Say, there is big body of smoke visible to you from a kilometer ahead of you. What is your guess? Yes, you guessed it right. There is a big fire there. Did you see the fire? Probably not. You guessed on the basis of smoke that there is a fire there. This is called indirect way of interpretations. A beautiful example of such a test is ELISA where we do not know if the interaction has taken place or not. However, we can see it indirectly. We will see that in the following paragraph which discusses ELISA.

Enzyme-Linked Immunosorbent Assay: This test is done by coating the ELISA plastic plates either with the antigen or the antibodies. Suppose we have Brucella antigen coated ELISA plates that can detect antibodies from the serum of a Brucella infected animal. You take the serum and put into the wells of ELISA plate. You allow time for antigen and antibody to interact and then wash these wells with some buffer to remove unbound antibodies. These bound antibodies are not visible at this step. So, how do we see this interaction? Well, we use another antibody that has been made against the antibody bound to the antigen. This special antibody has been tagged with an enzyme. After allowing time for interaction of this 2_{nd} antibody with the 1_{st} antibody, 2_{nd} antibody is also given a washing to remove unbound antibody molecules. Again, we cannot say if interaction has taken place or not. Here comes an indirect way of detecting this interaction. Remember that the 2_{nd} antibody has a molecule of an enzyme attached to it. Now, if we provide substrate to the enzyme in the wells, enzyme will convert the substrate into a color compound visible to the eyes. The color change can be measured by a spectrophotometer as well which gives a quantitative data

100. Explain types of conidiophores.

101. Explain types of algae.

- **Phaeophyta (Kelp):** Commonly known as brown algae, brownish in color, contains cellulose and alginic acid cell walls, are multicellular and contain chlorophyll *a* and *c*, xanthophylls. These store carbohydrates and are used for algin which is a thickener in ice creams. Can grow upto 20 cm in one day.
- **Rhodophyta:** Red Alage, these are reddish in color and contain cellulose in their cell walls. Most are multicellular and contain chlorophyll a and d, and phycobiliproteins. They store glucose polymer and used for agar and carrageenan (both are used as thickeners of foods).

- **Chlorophyta:** Green Algae: have all the features of red algae except that chlorophyll is of a and b types only. These algae are believed to have given rise to terrestrial plants.
- **Bacillariophyta:** Cell wall is composed of pectin and silica. They are unicellular and possess chlorophyll a and c, and carotene xanthophylls. These algae store oil and produce domoic acid with which human can be intoxicated by eating contaminated mussels.

  Diatoms: Come in beautiful colors and shapes.
- **Dinoflagellata:** These are unicellular algae collectively called as plankton (free floating organisms). These cells have cellulose in their cytoplasmic membrane which gives the membrane rigidity. They have chlorophyll *a* and *c*, and carotene. These algae store starch in them. Dinoflagellates in the genus Alexandrium produce neurotoxins (called saxitoxins) which can cause paralytic shellfish poisoning in humans when they consume contaminated mussels and clams. Blooms of Alexndrium gives ocean a deep red color commonly known as red tide.
- **Oomycota (Water molds):** They are basically decomposers of dead matter. Superficially, these resemble with fungi; however, these are motile with flagella and their cell wall is composed of cellulose not chitin. Their spores are called zoospores.
- Terrestrial members of water molds are also plant parasite. *Phytophthora infestans* was responsible for Irish potato blight during mid-1800s.

  P. ramorum causes sudden oak death

2 & 3 Marks:

6.
7. Write economical uses of fungi.
8. What is sporulation?
9. Differentiate between viable and direct count.
10. What is flagellum? Write its functions.
11. What is meant by gene transfer? Name the types.
12. Describe classification system proposed by Robert H Whittaker.
13. What is taxonomic hierarchy?
14. What is western blotting?
15. Enlist types of passive transport.
16. Name types of spiral bacteria.
17. Define passive movement.
18. What is function of lysozymes?

28. Write methods for viable count of bacterial colonies.

5 & 10 Marks Questions:

1. Explain characteristics of aerophiles.

3. Explain methods of microbial control in detail.
4.

6. Explain methods of microbial identifications.
7. How fungi can be identified by biochemical testing?
8. Label the diagram of Electron microscope with numbers mentioned.
9. Write functions of cell membrane of bacteria.
10. Write names of 5 types of bacteria with respect to temperature.
11. Write characteristics of metachromatic granules.
12. Explain five kingdom classification systems.
13. Explain layers of endospore.
14. Enlist types of bacteria on the basis of oxygen.
15. Name five types of microscopy.
16. Explain ELISA.

18. Explain types of algae.

1: What are chracteristics of Microaerophiles? 2mks

8:How bacteria are identify? Give detail of any two methods?5
9: How infectious diseases are transmitted?

1. Characteristics of magnetosomes.

1.what are flurochromes...
4.importance of fungi..
5.microbial cntrol
6.advantages of staining..
7.metachromatic granules..
9.transmissn of infection..
1 Micribial identifcattion method. 2 Transmission of microbes passive movemnt (2) flagellm and function (2) how disease can be transmitd (5) Any two microbial identification method (5) Magnetostome (5) immunity and its types in detail (10)

2) Function of lipid A in lipopolysaccrides.(2)
2 marks ques :
1.Glycoclyx, its composition and function
Characters tics of aerophiles (2)
1. L bacteria3marks

Write composition of alchic acid equation?(2)

2MARKS QUESTIONS

1. **Define magnification**

Magnification is the enlargement of an object its formula is Magnification of objective lens x magnification of ocular lens

2. **Define Buffer**

A buffer is the substance that maintains the desired pH in the solution or any medium.

3. **What is taxonomic hierarchy**

Microbes are placed in groups based on similarities that they share with each other. All organisms can be grouped into a series of subdivisions that make up the taxonomic hierarchy.

4. **Give two advantages of solid media in bacterial culturing.**
   - Pure culture is easy to obtain by streaking the organisms
   - Colony characters can be studied

5. **Physical Requirements of Media**

Physical: Requirements • Temperature • pH • Osmotic Pressure

6. **Passive movement**

The movement of molecules from high concentration to low concentration without use of ATP (energy) is called passive transport. Example, Simple Diffusion, Facilitated diffusion and Osmosis

7. **Flagellum and its function (2)**

Long filamentous structure that propels bacteria, the primary role of the flagellum is locomotion, but it also often has function as a sensory organelle, being sensitive to chemicals and temperatures outside the cell. **Example Chemotaxis –Phototaxis**

8. **Glycoclyx, its composition and function**

It is the outermost layer, viscous and gelatinous in nature that surrounds the cells. It is composed of polysaccharide and polypeptide or both. • Capsule: If glycol calyx is organized and firmly attached to the cell, it is called a capsule.– Plays important role in virulence (degree of pathogenicity)–Can also be a part of the vaccine against the bacteria to which it belongs.–Can be demonstrated by negative staining technique

9. **Characters tics of microaerophiles**

A microaerophile is a microorganism that requires oxygen to survive, but requires environments containing lower levels of oxygen than are present in the atmosphere (i.e. <21% O_2; typically 2–10% O_2). Many microaerophiles are also capnophiles, requiring an elevated concentration of carbon dioxide (e.g. 10% CO_2 in the case of *Campylobacter* species)

JOIN VU BIO ANIMALS

10. **Write composition of teichoic Acid?**

    Teichoic Acid Composition: - (glycerol or ribitol + Phosphate)

11. **Differ b/w chemotaxis and phototaxis**

    Phototaxis: - The movement of organism or its part towards the light whiles the movement of an organism towards specific stimulus or chemical substance called chemotaxis.

12. **What is western blotting**

    We can use this technique for the detection of antigen in the serum. Microbial proteins can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins

13. **Define virulence**

    The ability of an agent of infection to produce disease, the **virulence** of a microorganism is a measure of the severity of the disease it causes.

14. **Two teichoic acid present in bacterial cell wall.**

    1) Lipoteichoic Acid 2) Wallteichoic acid

15. **Define antiseptics**

    An **antiseptic** is a substance that stops or slows down the growth of microorganisms. Like Ethanol , hydrogen peroxide etc.

16. **What is high frequency combinations**

    If F factor gets integrated into the chromosome of the recipient cells, the bacterium becomes high frequency of recombination cell, it is a kind of conjugation.

17. **Differentiate between clone and strain**

    Clone: Population of cells derived from a single cell that is genetically identical. **Strain:** A genetic variant of a clone is called a strain

18. **Difference between positive sense and negative sense RNA**

    The main difference between positive and negative sense RNA virus is that positive sense RNA virus consists of viral mRNA that can be directly translated into proteins whereas negative sense RNA virus consists of viral RNA that is complementary to the viral mRNA.

19. **What is zoonosis?**

    Primarily an animal infection, but can be transmitted to humans. It can assume epidemic proportions in humans and is difficult to control

20. **Q. Write characteristics of Microaerophiles**

    1) Aerobes but O2 is required in low amounts. 2) Produce O2 toxic compounds if exposed to more O2

21. **Q Name two basic dyes (2)**

    Crystal violet • Methylene blue • Malachite green • Safranin

JOIN VU BIO ANIMALS

22. **Q Two methods for Viable counting (2)**

    1) Spread plate 2) Pour plate

23. **Q.4 what is microbiology**

    Studying microbes is microbiology

24. **Q. characteristics of Integral proteins.(2)**

    Imbedded in the membrane –Insoluble in water –Amphipathic in nature

25. **Q. Formula of magnification?**

    Magnification of objective lens x magnification of ocular lens

26. **Q. Substrate level phosphorylation 2**

    The Direct transfer of P to ADP

27. **Q. Macconky agar 3**

    Nutrient agar + Bile salt + CV • Selective & differential, Used for gut bacteria

28. **Q. what is virulence?2**

    Virulence: The ability of an agent of infection to produce disease. The ***virulence*** of a microorganism is a measure of the severity of the disease it causes.

29. **Q. Advantages of solid medium.2**

    Pure culture is easy to obtain by streaking the organisms • Colony characters can be studied

30. **Pasturization method**

    It is a method in which the liquid usually milk is pasteurized to Reduces spoilage organisms and pathogens

31. **Q. Two types of fermentation?**

    Lactic acid fermentation and alcoholic fermentation

32. **Q. define L-form bacteria 2m**

    L-form organisms are naturally found without the cell wall. In other words, normally, they have cell wall but under certain conditions, they may lack a cell wall.

33. **Q. Industrial microbiology**

    Use of microbes or their enzymes for large scale production of biomolecules

34. **Q. Give classes of media based on function 3m**

    Defined and complex media • General purpose media • Enriched media • Selective media • Differential media

Join VU BIO ANIMALS

3MARKS QUESTIONS

35. Describe transduction, conjugation transformation (3)

- **Transduction:-** Transferring of a gene from a bacterium to another bacterium via a virus is called transduction
- **Conjugation:-** Bacterial conjugation is the transfer of genetic material between bacterial cells by direct cell-to-cell contact or by a bridge-like connection between two cells
- **Transformation:-** Transformation is the process by which a DNA molecule is taken up from the external environment and incorporated into the genome of the recipient cell or uptake of naked DNA by microbes.

36. Economical use of fungi (3)

- Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production.
- Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO2 which expands the dough causing it to rise
- Yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from Aspergillus niger
- antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products
- Mushrooms can be cooked and eaten as a source of proteins for humans
- Entomophaga (kills gypsy moth), hence is used as a biological pest control

37. Difference between vertical and horizontal gene transfer (3)

- **Vertical Gene Transfer**: - This is a normal way of transferring genes from parents to offspring. This happens when a cell divides. Each daughter cell receives exactly what its parent cell has.
- **Horizontal Gene Transfer**: - When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer. This can happen between cells of the same species, or across different species of organisms.

38. Define Plasmogamy, Karyogamy and Meiosis (3)

- **Plasmogamy:** A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-).
- **Karyogamy**: The (+) and (-) nuclei fuse to form a diploid zygote nucleus. Sexual Spores
- **Meiosis:** The diploid nucleus gives rise to haploid nuclei (sexual spores).

39. What are interferons? give their functions

These are antiviral proteins, produced by macrophages and lymphocytes in response to viral infections. Fibroblasts can also secrete it. These proteins interfere with viral replication

Types of Interferon:

- Alpha: It is released by all infected cells.
- Beta: Also produced by infected cells.

Join VU BIO ANIMALS

- Gamma: This one is released by lymphocytes only. Lymphocytes belong to adaptive immune system (just remember that).

40. -L form bacteria

L-form organisms, are naturally found without the cell wall. In other words, normally, they have cell wall but under certain conditions, they may lack a cell wall..

41. Function of cell wall in bacteria(3)

1) Shape 2) Protection from osmotic lysis 3) Pathogenicity

42. Define Virus, virusoid, viroids(3)

- **VIRUSES:** They are acellular structures that consist of a protein capsid enclosing either DNA or RNA (but never both) and sometimes may contain an extra membrane called envelope
- **VIROIDS:** They are composed only of circular single stranded (ss) RNA
- **VIRUSOIDS:** They contain circular ssRNA that need helper viruses for replication and encapsidation. They are also called satellite viruses
- **PRIONS:** These are basically infectious proteins formed inside the body. They behave like organisms as they can be transmitted from one individual to another. Extremely resistant to heat.

43. DNA lateral transfer

Lateral gene transfer or Horizontal gene transfer: - When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer. This can happen between cells of the same species, or across different species of organisms.

44. Types of Spiral Bacteria?

Vibrio: curved rods. **Spirillum:** Helical but rigid **Spirochete:** Helical but flexible

45. Characteristic of bacterial colony. (3)

- **Forms of bacterial colonies** like Punctiform, circular, filamentous etc.
- **Elevation of bacterial colonies** like Flat, raised, convex etc.
- **Margins of Bacterial colonies** like entire look circular, undulate, curled etc.

46. Magnetosomes characteristics.

- Inclusions of iron oxide
- Surrounded by invaginations of plasma membrane
- Present in G negative bacteria

47. Functions of bacterial cell wall.3

1) Shape 2) Protection from osmotic lysis 3) Pathogenicity

48. what are the physical and chemical methods for making Smear on the slide

a. Physical method: Heat
b. Chemical Methods: Ethanol or formaldehyde

LONG QUESTION

Q.1 Epidemiology and its methods 5

ANS- The study of where and when diseases occur and how they are transmitted in a population

Descriptive Epidemiology: (recording the data about disease). It includes location and time of cases of disease, gender, health, age of patients etc. (Basically, frequency and distribution of risk factors in population are recorded). Prevalence and incidence of disease are typical examples. Such studies can be retrospective (after they occur) or prospective (before they occur). So, in summary, in descriptive epidemiology, information about an already existing disease is recorded.

Analytical Epidemiology: It relates to determining the cause of the disease. Analysis of data, mode of transmission of diseases and means to prevent diseases also come in this discipline. Such studies can be done with case.

Control methods in which factors that have preceded the disease are determined. A group of people who have the disease is compared with another group of people in the same location who are free of the disease.

A **Cohort method** tracks two groups forward from exposure to outcome. Cohort study compares the experience of a group exposed to the factor with another group not exposed to the factor.

Q-2 How infectious diseases transmit? Give detail of transition by direct contact, vector and vehicles 5

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example

Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.

Droplet Transmission: Mucus droplets are created when you sneeze for example. These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

What are Vectors? A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another.

Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another. Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another

Join VU BIO ANIMALS

Q-3 How fungi can be identified by biochemical testing,

Fungal biochemical tests are tests performed to identify fungi.

The following are some systems currently being used for the identification of yeasts:

- **Carbohydrate fermentation** studies: This tests whether a certain yeast ferments different carbohydrates.
- **Carbohydrate assimilation** studies: This tests whether yeast can aerobically utilize different carbon and nitrogen compounds with oxygen as their final electron acceptor.
- **API 20C System**: Freeze-dried sugars are placed into wells on a plastic strip, yeast basal medium suspended yeast is pipetted into wells, and incubated. Turbidity determines growth.
- **Uni-Yeast Tek System**: Identifies yeasts based on their ability to ferment certain sugars

Q-4 Explain types of algae

Phaeophyta (Kelp)

1) Brownish in color 2) Cellulose and alginic acid cell walls 3) Multicellular 4) Chlorophyll *a* and *c*, xanthophylls

5) Store carbohydrates 6) Harvested for algin

Rhodophyta

1) Reddish in color 2) Cellulose cell walls 3) Most are multicellular 4) Chlorophyll *a* and *d*, phycobiliproteins

5) Store glucose polymer 6) Harvested for agar and carrageenan

Chlorophyta

1) Green algae 2) Cellulose cell walls 3) Unicellular or multicellular 4) Chlorophyll *a* and *b*

5) Store glucose polymer 6) Believed to have given rise to terrestrial plants

Bacillariophyta

1) Brownish in color 2) Pectin and silica cell walls 3) Unicellular 4) Chlorophyll *a* and *c*, carotene, xanthophylls

5) Store oil 6) Fossilized diatoms formed oil 7) Produce domoic acid

Dinoflagellata

1) Brownish in color 2) Cellulose in plasma membrane 3) Unicellular
4) Chlorophyll *a* and *c*, carotene, xanthins 5) Store starch 6) Some are symbionts in marine animals
7) Neurotoxins cause paralytic shellfish poisoning

Q-5 Explain types of conidiophores

Conidiospores

- **Conidia:** in this form Chain of spores are formed at the end of a conidiphore for example Aspergillus
- **Arthroconidia:** in this form spores are formed by fragmentation of septate hypha for example Coccidioides immitus
- **Blastoconidia:** spores are formed in the result of budding of the parent cells for example Candida albicans
- **Chlamydioconidia:** spores formed by rounding and enlargement within a hyphal segment Example Candida albicans

Q-6 Explain ELISA (5)

Enzyme Linked Immunosorbent Assay –ELISA

- Some antigen antibody reactions are not visible to the eyes.
- Such reactions are visualized indirectly using various probes.
- Probes include enzymes, radioisotopes etc. Serological Testing
- Enzymes or radioisotopes are conjugated to antibodies
- Presence of enzymes detected by incubating with substrate that forms colored compounds

Q-7 Microbial identification method. Give detail of two. 5

1) **Enzyme Linked Immunosorbent Assay –ELISA**

- Some antigen antibody reactions are not visible to the eyes.
- Such reactions are visualized indirectly using various probes.
- Probes include enzymes, radioisotopes etc. Serological Testing
- Enzymes or radioisotopes are conjugated to antibodies
- Presence of enzymes detected by incubating with substrate that forms colored compounds

2) **Western Blotting**

Western Blotting: We can use this technique for the detection of antigen in the serum. Microbial proteins can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) is present on the gel. Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies

Join VU BIO ANIMALS

10 MARKS QUESTIONS

Q1. **Define immunity, difference between innate and adaptive immunity in Detail (10)**

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Susceptibility: lack of immunity which leads to easy establishment of disease.

Types of Immunity

1) **Innate (Natural):**

- **Present at birth:** One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.
- **Rapid to respond:** Because it is already there, it interacts with the microbes as soon as they enter the body.
- **Nonspecific**: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).
- Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.
- **First line of defense:** Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.
- **Second line of defense**: Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.

2) **Adaptive (Acquired): It is the third line of defense.** Acquired by experience, specific to an organism.

- **Specific:** It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.
- **Have memory**: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response. O
- **Because of memory**, body's immune response increase with every repeated exposure. It has a limit also!
- **Slower to respond:** For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.
- **Adaptive Immune Response** can further be divided into:
    - **Humoral immune response** (antibody response by B cells or lymphocytes)
    - **Cellular immune response** (T cells are made which kills microbial harboring cells).

Join VU BIO ANIMALS

MIDTERM 5 MARKS QUESTIONS INCLUDED IN FINAL TERM

Q.1 Write About the five layers of endospore?

- **Exosporium:** A thin delicate outermost covering of the spore
- **Spore coat:** 2nd layer underneath the exosporium. It is thick and composed of several protein layers.
- **Cortex:** It is the 3rd layer from outside in. It has peptidoglycan in it.
- **Spore cell wall or core wall**: Surrounds the protoplast or spore core
- **Spore core:** Contains nucleoid and ribosomes

Q.2 Function of cell membrane in bacteria?

1. Selective permeable barrier
2. Passive and Active Transport
3. Respiration in microbes (bacteria)
4. Photosynthesis in microbes
5. Lipid synthesis
6. Cell wall parts are transported by a molecule called bactoprenol which is present in the plasma membrane.
7. PM has many receptors in it with which various ligands can bind and initiate signals for gene expression etc

Q.3; Write names of kingdom as per five kingdom system of classification.

In 1969, five kingdom classification was proposed by Robert Whittaker as under.

1. **Kingdom Monera:-** The Kingdom Monera consists of prokaryotic, unicellular organisms. No nuclear membrane or membrane-bound organelles such as chloroplasts. Examples of Monera include Archaea and Bacteria.

   - **BACTERIA:** These are prokaryotes, usually single-celled organisms. Their cell wall is made of peptidoglycan. These are true bacteria

   - **ARCHEA:** These are also prokaryotes but their cell wall lacks peptidoglycan (may have special lipids

2. **PROTISTS:** This group includes algae, protozoa, slime molds and water molds.

   - **Algae**: These are eukaryotes. Their cell wall consists of cellulose. These are photosynthetic and produce oxygen

   - **Protozoa:** These are also eukaryotes, unicellular and have no cell wall. They move by pseudopods, flagella or cilia. Some are photosynthetic

   - **Slime Molds:** They are fungus like organisms at one time that may assume a protozoon like nature at another time. Basically, they are single celled but aggregate to form multicellular reproductive structures

JOIN VU BIO ANIMALS

- **Water Molds:** These are protists that grow on the surface of fresh water and humid soils. Also known as mildews and white rusts. They look like fungus

3. **FUNGI**: Usually unicellular (but also multicellular), their cell wall consists of chitin

4. **Plantae:** Multicellular eukaryotic and Autotrophs, photosynthesis. Like plants

5. **Animalia:** Multicellular eukaryotic and heterotrophs

Q.4; Characteristics of Metachromatic granules

1. Also called volutin, they stain red with certain dyes such as methylene blue. That is why they are called metachromatic (stain in different color as methylene blue gives blue color but the color on these granules is red).
2. Large inclusions
3. These inclusions contain inorganic phosphates. Inorganic phosphates are used up in ATP synthesis.
4. Characteristics of Corynebacterium diphtheria: This bacterium can be identified by the presence of these granules in it.

Q.5; Pili structure and its functions

Structure

1. These are hair-like structure composed of pilin, usually one to ten in number.
2. Longer than fimbriae

Function

3. Used for attachment to: • Host cells • Bacteria
4. Used for DNA transfer from on bacterium to another: Conjugation (Sex pili)
5. Also function in twitching Motility
6. Gliding Motility is also the function of the pili

Q.6; Name of types of microscopy

Types of Microscopes:

1. **Darkfield microscopy**
   - Uses a darkfield condenser having an opaque disk
   - Used for unstained specimens suspended in liquids
2. **Phase contrast microscopy**
   - Light rays can be in phase or
   - out of phase
3. **Fluorescent microscopy**
   - Uses fluorochoromes and UV light as illumination source
   - Fluorochoromes: substances that absorb short wavelength of light and emit longer wavelength (visible).
4. **Transmission electron microscopy**

Join VU BIO ANIMALS

- Electron microscope (EM) makes use of electron beam as an alternative to light rays
- It is used to analyses the internal structure of the organisms like cell organelles etc

5. **Scanning electron microscopy**
   - It is modified type of Electron microscopy in which electron beam doesn't pass through the specimen and the entire structure can be seen under this microscope

Q.7 Types of bacteria based on oxygen requirement for growth

Based on the use and concentration of Oxygen, microbes can be classified into five groups:

1. **Obligate Aerobes:** Oxygen must be present for their growth. These organisms do not grow if O2 is not present in the medium. If you test these organisms in a tube containing nutrient agar, the growth will remain confined to the surface only. **Mycobacterium tuberculosis** is an example of such an organism.
2. **Facultative Anaerobes:** These organisms prefer aerobic respiration; however, if oxygen is not available, they can use anaerobic respiration or fermentative modes for generating ATPs.
3. **Obligate Anaerobes:** Although, these organisms do not use oxygen, they cannot grow in the presence of oxygen. The reason is that oxygen creates toxic compounds which can kill these cells.
4. **Aerotolerant Aerobes:** These organisms do not use oxygen and also are not bothered by the presence of oxygen. In other words, they have a better system to dispose of toxic oxygen compounds. They will grow evenly throughout the tube of medium as seen in the accompanying diagram.
5. **Microaerophiles:** –Aerobes but O2 is required in low amounts. –Produce O2 toxic compounds if exposed to more O2 Examples Listeria, Vibrio

Q. 5 Classifications of bacteria on the basis of optimal temperature?

Organisms are basically classified into three groups based on the temp requirements:

1. **Psychrophiles:** These are further divided into strict psychrophiles and psychrotrophs:
   - Psychrotrophs: Cold loving: 15 0C
   - Psychrotrophs: Optimum temp is 20-30 0C. Food spoilage bacteria that can spoil food during refrigeration.
2. **Mesophiles: 25 – 40 0C:** These are the ones that cause diseases in animals and humans.
   - Moderate temp loving organisms
   - Optimum: 37 0C as this is the body temperature of humans and animals.
3. **Thermophiles: 50 – 60 0C**
   - Heat loving
   - Important in organic compost piles
   - These are further divided into hyper thermophiles that grow optimally at 80 0C. They live in hot springs.

Q. Name 5 types of microbiology.5

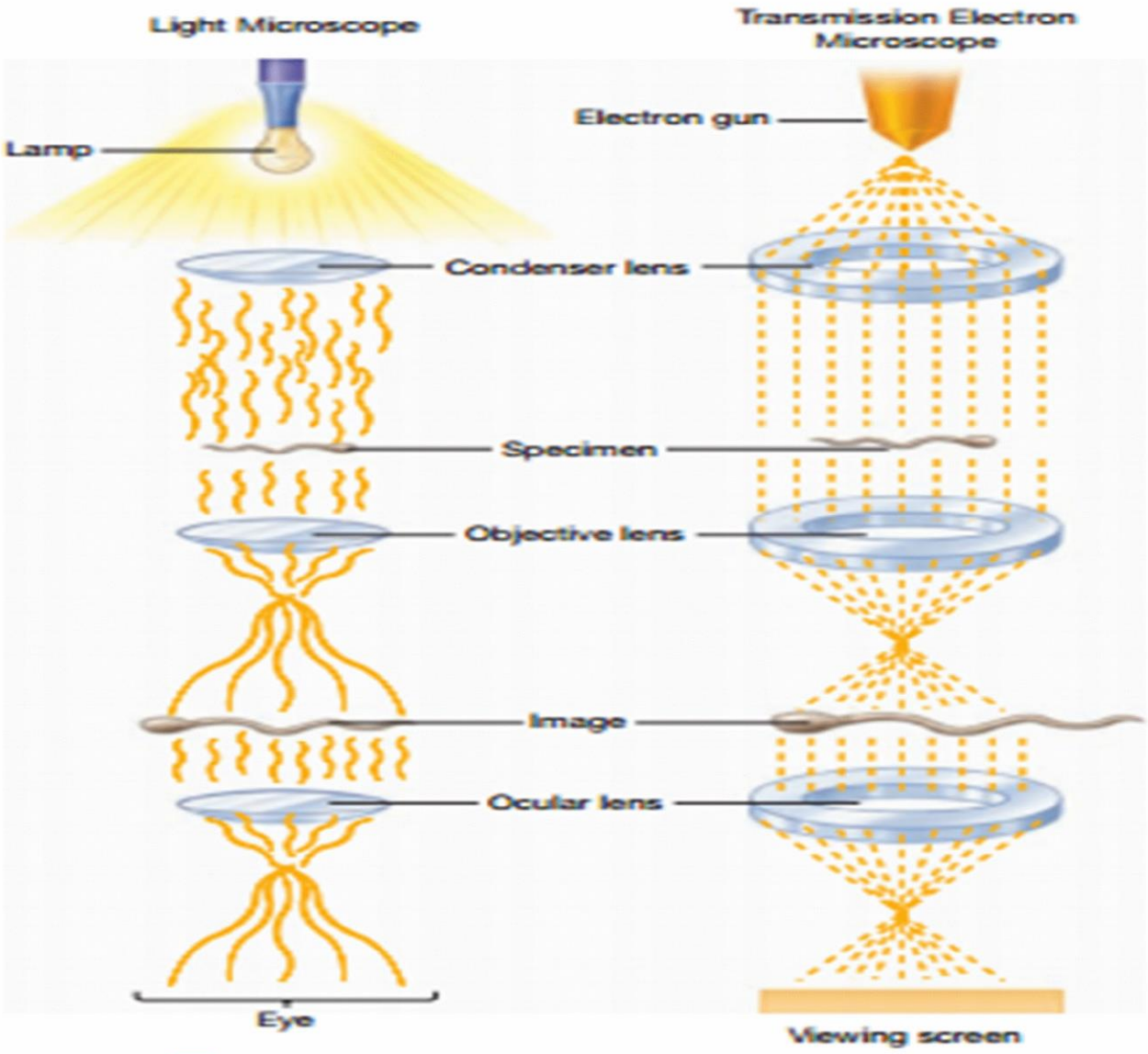
1) Bacteriology 2) Mycology 3) Medical Biology 4) Immunology 5) Food microbiology

Join VU BIO ANIMALS

Q. Classes of media based (5)

Defined and complex media • General purpose media • Enriched media • Selective media • Differential media

Q. Label Electron Microscope 5



Join